Project Design Phase
Problem – Solution Fit Template

Date	08 October 2026
Team ID	SWTID-2026-2540
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	2 Marks

Problem – Solution Fit Template:

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why

Purpose:

- ❑ Solve complex problems in a way that fits the state of your customers.
- ❑ Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.
- ❑ Sharpen your communication and marketing strategy with the right triggers and messaging.
- ❑ Increase touch-points with your company by finding the right problem-behavior fit and building trust by solving frequent annoyances, or urgent or costly problems.
- ❑ **Understand the existing situation in order to improve it for your target group.**

Problem – Solution Fit: Script-Controlled ACL

Restricting Institution Details record access by branch value and role in ServiceNow

1. CUSTOMER SEGMENTS [CS]

- Institution users who access student/institution records in ServiceNow.
- EEE branch users assigned the bb1 role.
- Administrators who require unrestricted access for management and testing.

6. CUSTOMER CONSTRAINTS [CC]

- Access must depend on both user role and the Branch field value.
- Unauthorized users must not view protected records.
- Admin users must retain full access.
- Security must apply consistently to list and form access.

5. AVAILABLE SOLUTIONS [AS]

- ServiceNow record ACLs for read, create, write and delete operations.
- Custom roles bb1, bb2, bb3 and bb4.
- Advanced read ACL script using gs.hasRole().
- Data condition: Branch is EEE.

2. JOBS-TO-BE-DONE / PROBLEMS [JP]

- Prevent users from seeing institution records outside their authorized branch.
- Allow EEE users to work only with EEE records.
- Control create, edit and delete capabilities independently.
- Avoid relying only on UI visibility for security.

9. PROBLEM ROOT CAUSE [RC]

- A custom table does not automatically enforce branch-based record visibility.
- Roles alone identify permissions but do not evaluate each record's Branch value.
- Without ACL rules, sensitive records may be exposed to users who should not access them.

7. BEHAVIOR [BE]

- bb1 user opens the Institution Details list and sees only EEE records.
- bb2 enables creation, bb3 enables editing and bb4 enables deletion.
- Users lacking the required role are denied the corresponding operation.
- Admin can access all records regardless of branch.

3. TRIGGERS [TR]

- A user attempts to open, create, modify or delete an Institution Details record.
- ServiceNow evaluates the matching ACL before permitting the operation.

4. CUSTOMER EFFECT / AFTER [EM]

- Authorized users get only the access needed for their role.
- Sensitive branch records remain protected from unauthorized viewing or modification.

10. YOUR SOLUTION [SL]

- Create u_institution_details with Branch choices ECE, EEE and CSE.
- Create roles bb1–bb4 and assign them according to required permissions.
- READ ACL: require bb1, condition Branch = EEE, and advanced script allowing admin/bb1.
- CREATE ACL: bb2; WRITE ACL: bb3; DELETE ACL: bb4.
- Verify security by impersonating users and comparing access with admin.

8. CHANNELS & BEHAVIORS [CH]

- ServiceNow Application Navigator and Institution Details list.
- Record forms and list actions such as New, Edit and Delete.
- Impersonation is used to validate each permission combination.
- ACL evaluation provides server-side enforcement across supported interfaces.